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ASSIGNMENT 9

Problem Statement: Write a program in C to check whether the given number is an Armstrong number or not. An Armstrong no. is an integer with 3 digits such that the sum of cubes of its digits is equal to number itself.

Objective:
- To understand the concept of Armstrong number.
- To practice arithmetic operations in C.
- To develop logical thinking using loops & conditional statements.

Theory: An Armstrong number is a number that is equal to the sum of cubes of its digits. For a three-digit number etc, it is an Armstrong num. if -
$a^3+b^3+c^3 = abc$.

Ex: 371 is an Armstrong number because:
$3^3+7^3+1^3 = 27+343+1 = 371$.

Platform: Linux

Algorithm:

Step 1: START.
Step 2: Declare variables num, temp, digit, sum=0.
Step 3: Read the number num from the user
Step 4: Store the value of num in temp
Step 5: Repeat while temp ≠ 0
- Extract digit: digit = temp % 10
- Add cube of digit to sum: sum = sum + (digit x digit x digit)

- Remove last digit : temp = temp/10

Step6: Compare Sum with ~~sum~~ num
- If sum == num, print "Armstrong No."
- Else print "Not an Armstrong No."

Step7: STOP.

C Program:

```
# Include<stdio.h>
int main() {
    int num, temp, digit;
    int sum=0;
    printf("Enter 3-digit no:");
    scanf("%d", &num);
    temp= num;
    while (temp!=0)
    {
        digit= temp % 10;
        sum= sum+ (digit * digit * digit);
        temp= temp /10;
    }
    if (sum == num)
        printf("%d is an Armstrong no.", num);
    else
        printf("%d is not an Armstrong no.",
            num);
    return 0;
}
```

Input:
Enter a three-digit number.
Ex Input: 371

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Output:
Ex. output:
371 is an Armstrong number.

Conclusion: Thus, we have successfully written a C program to check whether a given number is an Armstrong number using loops & arithmetic operations.

FAQs:

1. What is an Armstrong number?
A An Armstrong number is a three-digit number in which the sum of cubes of its digits is equal to number itself.
Ex: $371 = 3^3 + 7^3 + 1^3 = 371$

2. Why is the Armstrong number concept important in programming?
A. The Armstrong number concept helps in understanding basic programming logic such as digit extraction, use of loops, arithmetic operations & condition statements. It improves problem-solving skills & logical.

thinking.

3. How does this program work step by step?

A. ∘ The program reads a three-digit number from the user.
∘ The number is stored in a temporary variable.
∘ Using a loop, each digit is extracted using the modulus operation (%)
∘ The cube of each digit is calculated & added to a sum.